Lending Game: Results in Round 1

A. Simulation of the Market

Recall that for each loan application, your team is in competition with two other teams, which have different information sets than yours. The identity of the two teams that compete with you are different for each loan application. For example, if you are a type 1 lender (i.e., you have variable 'signal1' for new applications), for each loan application one of your competitor is a type 2 lender (i.e., has variable 'signal2') and the other is a type 3 lender (i.e., has variable 'signal3'). Because four teams are type 2 lenders (call them 2a, 2b, 2c and 2d) and four teams are type 3 lenders (call them 3a, 3b, 3c and 3d), we simulated the market by randomizing the identify of your competitors for each loan application over the sixteen (four time four) possible situations (your competitors may be 2a and 3a; 2a and 3b; 2a and 3c; etc.). We also randomized loan applicants' types, that is, the lender for which each loan applicant is ready to pay 2% extra to borrow from this lender.

B. Diagnostic Data

The detailed outcome of the market is in the dataset Round1_Diagnostic_xx.csv. It contains the 100,000 loan applications. The variables in the dataset are (if you are a type 1 lender):

- id: Loan identifier.
- own: Interest rate offered by your team.
- competing2: Interest rate offered by lender 2. Missing interest rate means no offer.
- competing3: Interest rate offered by lender 3.
- borrowertype: =1,2,3 indicates the lender for which the borrower is ready to pay 2% extra.
- accepted: =1 if the borrower takes your loan offer; =0 if the borrower does not take your loan offer. Note that you can reconstruct this variable using the information about the three interest rates and the borrower's type.
- default: =1 if the borrower defaults on the loan; =0 if the borrower repays the loan.
- profit: Your profit on this loan. The profit if your loan offer is accepted is equal to interest rate times 10,000 if the borrower repays, and to negative 10,000 if the borrower defaults.

C. League Table

The table below shows the market shares and the profits and losses (P&L) of all the teams participating to the game, which includes nine teams of students (three type-1 lenders, three type-2 lenders, and three type-3 lenders,) and three teams simulated using an LLM (one of each lender type). The market share for each team is calculated as the market share on all loans for which it competes. The P&L for each team is the profit summed over all the loans made by the team.

	Market share	P&L
LookingForAnInternshipInFinance Enterprises	37.5%	73,310,618
HECredit	38.4%	62,511,118
The Walking Debt	41%	53,883,986
Ponzi & Co	55.6%	35,004,136
Happiness in the Household	49.2%	30,908,296
Goldman Stanley	51%	28,677,721
Lehman Sisters	45.5%	23,423,067
Only God Can Trust Us	40.6%	10,744,247
ChatGPT Lender1	47.9%	5,379,147
ChatGPT Lender2	48.3%	4,798,398
ChatGPT Lender3	47.8%	4,356,905
Loan and Behold	26.3%	3,116,002

D. Second Round

All teams earned profits and most outperformed the LLM. Congratulations! This is quite an achievement because this business game is quite challenging.

Your task now is to use the diagnostic data to review your strategy and make offers to a new batch of loans. See the guidelines of the game for the detailed instructions. The simulation of profits in round 2 will be done in exactly the same way as in round 1. Good luck ☺